

Forecasting U.S. Mortgage Rates: Can Homebuyers Expect Relief Amid Recession Fears?

26S_CIS4680.01-1 Time Series Project | Executive Summary Report

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Problem Statement

The 30-year fixed mortgage rate is the most important number for anyone buying a home. It sets the monthly payment, decides who qualifies, and shapes what people can afford. Right now the backdrop is tense: oil is above \$100 a barrel, there is conflict in Iran, and betting markets put U.S. recession odds at 26%. Buyers are asking a simple question: wait for rates to drop, or buy now? This project answers it using 55 years of mortgage rate history and a 12-month forecast.

Approach

I pulled every weekly mortgage rate from April 1971 to May 2026 from the Federal Reserve and averaged them by month, giving 661 months of history. I held out the most recent 12 months as a test to see how well each model would have predicted what actually happened. I then trained four forecasting approaches on the rest of the data: Holt-Winters smoothing (default and tuned), SARIMA, and Prophet. These cover the main families of time series forecasting, Holt-Winters for trend-and-seasonality smoothing, SARIMA for autocorrelation-based prediction, and Prophet for changepoint-aware modeling. I picked the one closest to reality on the test year and used it to forecast the next 12 months.

Insights:

One model clearly won. Table 1 shows the tuned model predicted rates with 93.5% accuracy on the test year, an average miss of just 0.41 percentage points. The others were noticeably weaker, especially Prophet, which was off by nearly a full point. Figure 2 makes this obvious: the winning model (blue dashed) hugs the actual rates (green) closely while the others drift away. Figure 1 shows the 12-month forecast: rates stay flat in a band from 6.18% (Feb 2027) to 6.36% (Jun 2026), ending at 6.35% in May 2027, unchanged from today's 6.37%.

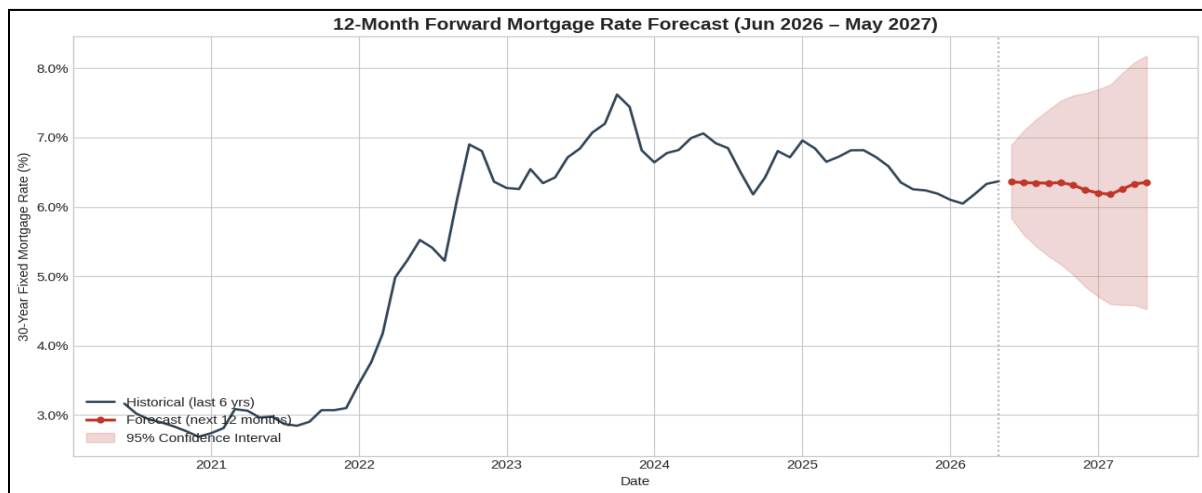


Figure 1. Forecast for the next 12 months. The red line is the projected rate; the shaded band is the range of plausible outcomes.

Table 1. How each model performed when tested against the last 12 months of real data.

Model	Average Error	Accuracy	Result
Holt-Winters (tuned)	0.41 pts	93.5%	Winner
SARIMA	0.46 pts	92.6%	Runner-up
Holt-Winters (default)	0.47 pts	92.5%	Baseline
Prophet	0.72 pts	88.7%	Weakest

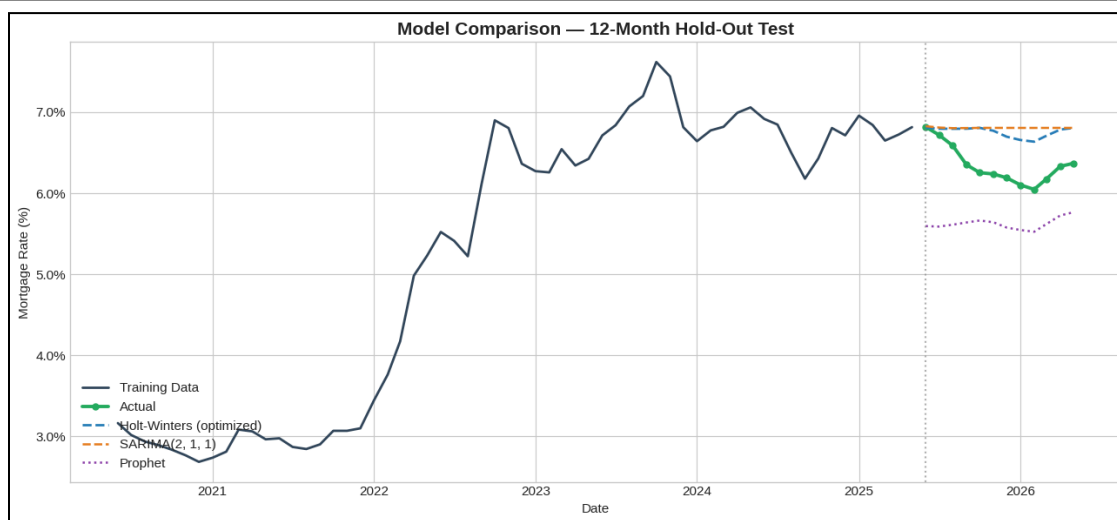


Figure 2. Each model's predictions (dashed lines) compared to what actually happened (green).
The closest tracker is the winner.

Interpretation:

The forecast is flat, and that is the point. Figure 1 shows the most likely path is sideways. The shaded band widens with time, an honest signal that uncertainty grows, but the central line stays near today's level. Nothing in 55 years of history points to a meaningful drop.

The model cannot see shocks coming. A recession would push the Fed to cut, dragging rates lower. An energy crisis would push them higher through inflation. Figure 1 captures neither; it assumes the future looks like the past. The confidence band partially reflects this risk, but a true regime shift would fall outside it.

Two scenarios worth pricing. If a recession arrives, rates could fall to the high 4s or low 5s, but that relief comes with job losses and tighter lending. If inflation reaccelerates, rates push to 7-8%, late-2023 territory. Both tails are plausible; neither is the central case.

Recommendation: Do not wait. If you are a homebuyer with stable income and a long holding period, the data does not support waiting for relief. Twelve months from now, rates are most likely where they are today. The higher-leverage moves are improving your credit, increasing your down payment, or buying in a market where prices have softened, all of which lower your monthly payment far more reliably than waiting for a rate cut the data does not predict.

Where this analysis falls short is this forecast looks only at mortgage rates in isolation. A stronger version would add the 10-Year Treasury yield, the Fed's policy rate, and inflation indicators, letting the model respond to macro shifts rather than just extrapolating the past. Future iterations should add those signals and re-run the forecast quarterly.